CHAPTER 1

Activities & Lab Packet

The Universe & Our Solar System

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 1. Works with both the Standard and Easy Read versions of the textbook.

📘 Georgia Standard S6E1

WHAT'S INSIDE

1 Pocket Solar System Lab

LAB

OBJECTIVE
Model the relative distances of the planets from the Sun and discover how empty the solar system really is.

TIME
40–50 minutes

STANDARDS
S6E1.b, S6E1.c

Materials (per group of 2–3)

- 1 strip of register tape or adding-machine paper, about 1 meter long (a cut strip of butcher paper works too)
- Markers or colored pencils
- Meter stick or ruler
- Planet distance card (bottom of the data sheet) and student data sheet (next page)

Setup

Cut one paper strip per group before class. Don't tell students where the planets will fall on the strip — most will guess the planets are evenly spaced. The "wow moment" is discovering that the four rocky planets are crammed into the first tenth of the strip while the outer planets are spread far apart.

Student Instructions

1. Label one end of your strip **SUN** and the other end **NEPTUNE**.
2. **Before measuring**, lightly pencil in where you *predict* the other 6 planets will go.
3. Fold the strip in half. Crease it. Open it up — mark the halfway crease **URANUS**.
4. Fold the SUN end to Uranus. The new crease is **SATURN**.
5. Fold the SUN end to Saturn. The new crease is **JUPITER**.
6. Fold the SUN end to Jupiter. This crease is the **ASTEROID BELT**. Notice: ALL four rocky planets must fit between here and the Sun!
7. Fold the SUN end to the asteroid belt: mark **MARS**. Fold to Mars: mark **EARTH**. Fold the Sun-to-Earth section in half: mark **VENUS**; **MERCURY** goes halfway between the Sun and Venus.
8. Compare the real positions to your predictions and complete the data sheet.

⚠️ SAFETY & TEACHER NOTES

Common confusion: Students mix up *distance* models and *size* models. This strip models distance only — at this scale, every planet would be smaller than a grain of sand!

Discussion starter at end: "Why do you think the rocky planets are bunched close to the Sun while the giants are spread out?"

Extension: Walk it outside! 10 big steps = Mercury's orbit, then have students compute the rest.

ANSWER KEY (data sheet)

1) Inner/rocky planets (Mercury, Venus, Earth, Mars) are squeezed into the first ~10% of the strip | 2) The gaps grow much larger as you move outward | 3) Most predictions space planets evenly — real data shows they are not | 4) Mostly empty space | Conclusion: a good model shows relative distance, but no model on paper can show distance AND size at once.

Pocket Solar System — Data Sheet

Name: ______________________ Period: __________

Planet	My prediction (where I guessed it would be)	Actual position (close to Sun? middle? far?)
Mercury		
Venus		
Earth		
Mars		
Jupiter		
Saturn		
Uranus		
Neptune		

1. What surprised you about where the rocky planets ended up?

____

____

2. What happens to the spacing between planets as you move away from the Sun?

____

____

3. How did your prediction compare to the real model?

____

____

4. Based on your strip, what is MOST of the solar system made of?

____

Conclusion: What does this model show well? What can it NOT show?

____

____

____

2 Orbit in a Bucket — Gravity & Inertia Lab

LAB

OBJECTIVE	TIME	STANDARDS
Model how gravity and inertia work together to keep planets in orbit.	30–40 minutes	S6E1.d

Materials (per group of 3–4)

- 1 sturdy string, about 1 meter long, tied securely to a tennis ball, beanbag, or rubber stopper (no hard or heavy objects)
- Open outdoor space or gym (each group needs a 4-meter "safety circle")
- Optional station 2: a marble and a round cake pan or pie tin
- Student data sheet (next page)

Setup

This works best outside or in a gym. Mark a spinning spot for each group with chalk or cones, far apart. Demonstrate the swing once yourself first: swing the ball in a slow, level circle overhead or at your side, then release and let the class watch the straight-line path it takes.

Student Instructions

1. **Station 1 — Ball on a string.** One partner swings the ball in a slow, steady circle. The string is **gravity** (pulling inward); the ball's motion is **inertia** (wanting to go straight).
2. While the ball circles, observers answer: what direction is the string pulling? What shape is the path?
3. On the teacher's signal, the swinger **releases the string** (away from people!). Watch carefully: what path does the ball take the instant "gravity" disappears?
4. Repeat so every group member sees a release. Record observations on the data sheet.
5. **Station 2 — Marble in a pan (optional).** Roll a marble around the inside wall of the pan. The wall acts like gravity. Tilt the pan so the marble escapes — watch its straight-line exit.

⚠️ SAFETY & TEACHER NOTES

Safety: Soft objects only, releases aimed at open space, and everyone outside the safety circle except the swinger. Goggles if indoors.

Key question to push: "The ball flew in a straight line, not a spiral. What does that tell you about what inertia does?"

Common confusion: Students often think objects keep curving after release. Newton (and the tennis ball) say no — with no inward pull, motion is a straight line.

ANSWER KEY (data sheet)

1) Inward, toward the hand (center) | 2) A circle/oval (orbit) | 3) A straight line | 4) String = gravity; moving ball = planet + inertia; hand = the Sun | 5) Earth would fly off in a straight line into space | Conclusion: orbit = gravity pulling inward + inertia moving straight ahead, balanced together.

Orbit in a Bucket — Data Sheet

Name: ______________________ Period: __________

1. While the ball is circling, which direction does the string pull it?

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2. What shape is the ball's path while the string holds?

---

3. Draw and describe the ball's path the moment the string is released:

Sketch (before release — label gravity & inertia)	Sketch (after release)

4. Match the model to the real solar system:

In our model…	…represents what in space?
The string	
The moving ball	
The hand at the center	

5. Predict: If the Sun's gravity suddenly disappeared, what would Earth's path look like?

Conclusion: Using the words GRAVITY and INERTIA, explain why planets orbit instead of flying away or crashing into the Sun.

3 Crater Creator — Impacts & Meteorites

LAB

OBJECTIVE
Model meteorite impacts and test how drop height (speed) and size affect craters.

TIME
40–50 minutes

STANDARDS
S6E1.e

Materials (per group of 3–4)

- Aluminum pan or shallow tub with 3–4 cm of flour
- Thin layer of cocoa powder or colored sprinkles sifted on top (makes ejecta visible)
- 3 "meteorites" of different sizes: e.g., a marble, a golf ball, a large bouncy ball
- Meter stick, ruler for measuring craters, newspaper or tablecloth for the floor

Setup

Place pans on newspaper — flour travels! Review vocabulary first: the rock in space is a **meteoroid**, the glowing streak is a **meteor**, and the object that lands is a **meteorite**. Your students are about to make meteorites.

Student Instructions

1. Smooth the flour and dust the surface with cocoa. This dark layer is the "crust."
2. **Test 1 — Height:** Drop the SAME ball from 30 cm, 60 cm, and 1 m. After each drop, measure the crater's width in cm, record it, then carefully lift the ball straight out.
3. **Test 2 — Size:** From the SAME height (60 cm), drop the small, medium, and large balls. Measure and record each crater.
4. Look closely at one crater: find the **rim** (raised edge) and the **rays/ejecta** (light flour splashed outward over the dark crust).
5. Re-smooth the flour between tests. Complete the data sheet.

⚠️ SAFETY & TEACHER NOTES

Safety: No throwing — drops only. Wipe up flour spills immediately (slippery). Check for gluten-allergy concerns; salt or sand works as a substitute.

Discussion starter at end: "The Moon is covered in craters but Earth shows very few. What might erase craters here?" (Weather, water, plants, plate movement — and our atmosphere burns small rocks up as meteors first!)

Common confusion: Bigger crater ≠ only bigger rock. Speed (drop height) matters too — that's why both tests exist.

ANSWER KEY (data sheet)

1–2) Craters get wider as height (speed) increases and as ball size increases | 3) Higher drop = faster impact = more energy | 4) meteoroid → meteor → meteorite | 5) Earth's atmosphere burns most up (meteors); weather and water erase old craters | Conclusion: impact energy (from speed and size) controls crater size.

Crater Creator — Data Sheet

Name: ______________________________ Period: ______________

Test 1 — Same ball, different heights

Drop height	Crater width (cm)	What the ejecta (splash) looked like
30 cm		
60 cm		
1 meter		

Test 2 — Same height (60 cm), different balls

Ball	Crater width (cm)	Observations
Small		
Medium		
Large		

1. How did drop height change the crater?

____

2. How did ball size change the crater?

____

3. Why does a higher drop make a bigger crater? (Hint: think speed and energy.)

____

____

4. Put the three names in order for one space rock's journey to the ground:

________________________ (in space) → ________________________ (glowing in the sky) →
________________________ (on the ground)

5. The Moon is covered in craters, but Earth isn't. Give one reason why.

____

____

4 Geocentric vs. Heliocentric — Evidence Sort

WORKSHEET

OBJECTIVE
Compare the two models of the solar system and explain why the model changed.

TIME
20 minutes

STANDARDS
S6E1.a

Name: ______________________________ Period: ______________

Part A. Write **GEO** (geocentric) or **HELIO** (heliocentric) next to each statement:

1. _______ Earth sits still at the center of everything.
2. _______ The Sun is at the center; planets orbit around it.
3. _______ This is the model supported by modern evidence.
4. _______ This model matched what people saw with just their eyes.
5. _______ Proposed by Nicolaus Copernicus.
6. _______ "Geo" is the root word in this model's name, meaning "Earth."

Part B. Short answers:

7. What did Galileo see through his telescope, and why did it damage the geocentric model?

____

____

8. Finish this sentence: *Scientific models change when…*

____

9. Pluto being reclassified as a dwarf planet in 2006 is another example of science changing. Was that a weakness or a strength of science? Explain.

____

____

____

ANSWER KEY

1) GEO 2) HELIO 3) HELIO 4) GEO 5) HELIO 6) GEO | 7) Moons orbiting Jupiter — proof that not everything circles Earth | 8) …new evidence appears / better tools give better information | 9) Strength — scientists updated the definition when they agreed on clearer rules; science improves itself.

5 Planet Data Detective

WORKSHEET

OBJECTIVE
Analyze and interpret planet data to compare size, distance, and ability to support life.

TIME
20–25 minutes

STANDARDS
S6E1.c

Name: ______________________________ Period: ______________

Use the data table to answer the questions.

Planet	Type	Size vs. Earth	Position from Sun	Could it support life?
Mercury	Rocky	Much smaller	1st	No — broiling days, freezing nights, no air
Venus	Rocky	About the same	2nd	No — hot enough to melt lead
Earth	Rocky	— standard —	3rd	YES — liquid water, breathable air
Mars	Rocky	About half	4th	Not now — too cold, thin air
Jupiter	Gas giant	Much larger	5th	No — no solid surface
Saturn	Gas giant	Much larger	6th	No — gas planet, extremely cold
Uranus	Ice giant	Larger	7th	No — freezing ice giant
Neptune	Ice giant	Larger	8th	No — coldest and most distant

1. What pattern do you see in WHERE the rocky planets are compared to the gas/ice giants?

   ____

2. Which planet is closest to Earth's size? ____________________

3. Why can't Jupiter support life, even if it were warmer?

   ____

4. Mars and Venus are Earth's neighbors. Give one reason EACH can't support life now:
   Venus: ____
   Mars: ____

5. Earth orbits in the "habitable zone." What does that distance make possible that life needs?

   ____

ANSWER KEY

1) Rocky planets are the 4 closest to the Sun; giants are the 4 farthest | 2) Venus | 3) It has no solid surface — it's made of gas | 4) Venus: crushing/poisonous atmosphere traps deadly heat; Mars: too cold with air too thin | 5) Liquid water (not too hot, not too cold).

6 Comet, Asteroid, or Meteoroid?

WORKSHEET

OBJECTIVE
Classify small solar-system objects by composition, location, and features.

TIME
15 minutes

STANDARDS
S6E1.e

Name: ______________________________ Period: ______________

Part A. Write **C** (comet), **A** (asteroid), or **M** (meteoroid):

____ 1. A "dirty snowball" of ice, dust & rock

____ 2. Mostly found between Mars and Jupiter

____ 3. Grows a glowing tail near the Sun

____ 4. Small rock that may become a "shooting star"

____ 5. Made of rock and metal; early solar system leftover

____ 6. Found at the far edges of the solar system

____ 7. Often a chip broken off an asteroid or comet

____ 8. Most orbit inside a "belt"

Part B. One rock, three names. Fill in the blanks with *meteoroid, meteor,* or *meteorite*:

9. A small rock drifting through space is a ______________________.

10. It streaks into Earth's atmosphere and glows — now we call the streak a ______________________.

11. A chunk survives and thuds into a field. The rock on the ground is a ______________________.

12. Memory trick: "-ite" means it ____.

Part C. 13. A comet's tail only appears during part of its orbit. When — and why?

____

____

14. Where is the asteroid belt, and what does it divide?

____

____

ANSWER KEY

1) C 2) A 3) C 4) M 5) A 6) C 7) M 8) A | 9) meteoroid 10) meteor 11) meteorite 12) hit the ground | 13) Near the Sun — the Sun's heat melts/vaporizes the ice, creating a glowing tail | 14) Between Mars and Jupiter; it divides the small rocky planets from the giant planets.

7 Writing Prompts

WRITING

OBJECTIVE	TIME	STANDARDS
Use scientific vocabulary to explain concepts in writing.	15–30 minutes each	Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Cosmic Address.** Write your full "cosmic address," starting with your street and ending with the universe. Then explain which level is the biggest jump in size, in your opinion.

2. **A Day as a Comet.** Pretend you are a comet at the cold, far edge of the solar system. Tell the story of your trip toward the Sun. What happens to you as you get closer? What do people on Earth see?

3. **Pick a Planet.** Choose your favorite planet (not Earth). Describe what type it is, where it is from the Sun, and the coolest fact about it.

Level 2 — Intermediate (3 paragraphs)

4. **The Lost String.** Use the ball-on-a-string demo to explain why planets orbit the Sun. Describe what each part of the demo represents — and what would happen to Earth if the Sun's gravity "let go."

5. **Real Estate Agent for Mars.** A company wants to sell vacation homes on Mars. Write an honest description: what makes Mars interesting, and which problems would make living there nearly impossible right now?

6. **Three Rocks Walk Into the Sky.** Explain the difference between a meteoroid, a meteor, and a meteorite by following ONE space rock on its journey from space to the ground.

Level 3 — Advanced (essay)

7. **Galileo on Trial.** Imagine you are Galileo defending the heliocentric model to people who believe Earth is the center of everything. Use specific evidence (like the moons of Jupiter) and explain why new evidence should change a scientific model — even a very old, popular one.

8. **The "Just Right" Planet.** Scientists searching for life on other worlds look for planets in the habitable zone. Construct an argument explaining why liquid water is the #1 thing they hunt for, using Venus, Earth, and Mars as your three pieces of evidence.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE
Practice scientific thinking through verbal reasoning.

USE
Warm-ups, exit tickets, Socratic seminar

STANDARDS
Cross-cutting

📘 **TEACHER NOTES**

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. The Sun looks like it moves across our sky every day. Why was geocentrism such an easy idea to believe?
2. Is the Sun a star, a planet, or something else? How do you know?
3. Which is bigger: a galaxy or a solar system? How many of one fits in the other?
4. If you see a "shooting star" tonight, did you see a star? What did you actually see?
5. Pluto used to be a planet and now it isn't. Did Pluto change, or did something else change?

Deeper Discussions (5–15 min each)

6. Copernicus and Galileo were told they were wrong by almost everyone. What should a scientist do when their evidence disagrees with what most people believe?
7. A planet's orbit needs BOTH gravity and inertia. What would the solar system look like with gravity but no inertia? With inertia but no gravity?
8. Venus is about Earth's size and is our neighbor — yet it's totally dead, while Earth is full of life. What does that tell you about how "picky" life is about conditions?
9. Astronomers keep finding planets around other stars. If we found one in its star's habitable zone, what ELSE would you want to know before saying it could support life?
10. The textbook compares the universe to "the whole planet Earth" while our solar system is just "your neighborhood." Why do scientists use comparisons like this instead of just giving the real distances?

9 Planet Travel Brochure

GROUP PROJECT

OBJECTIVE	TIME	STANDARDS
Research a planet and "sell" a trip there using accurate science.	2–3 class periods	S6E1.c

Project Description

Each pair or small group becomes a travel agency for ONE planet (or a dwarf planet/moon for a challenge). They create a tri-fold travel brochure that convinces tourists to visit — while being scientifically honest about the dangers. Finished brochures are displayed in Sun-to-Neptune order to build a classroom "travel wall" of the solar system.

Brochure Requirements

- **Front panel:** Planet name, a catchy slogan, and an illustration
- **Fast Facts box:** type (rocky / gas giant / ice giant), position from the Sun, size vs. Earth, and one record it holds (hottest, largest, windiest…)
- **"What to See" section:** at least 2 real features (Great Red Spot, Saturn's rings, Mars's giant volcano…)
- **"Travel Warnings" section:** at least 3 honest, scientific reasons the trip is dangerous (temperature, atmosphere, no surface to land on…)
- **Packing list:** 3 items a visitor would truly need, each with a one-sentence scientific reason

Assignment Ideas

Easier (more familiar info)	Medium	Challenging
Mars, Saturn, Jupiter, Earth's Moon	Mercury, Venus, Neptune	Uranus, Pluto (dwarf planet), Europa or Titan (moons)

Presentation Day

Each agency gives a 1–2 minute "sales pitch." The class votes on awards: *Most Convincing, Most Scientific, Best Warning Label.* Audience members jot one fact they learned about each planet on a "Boarding Pass" sheet.

📘 TEACHER NOTES

Differentiation: Assign planets by readiness — Mars and Saturn have the most accessible kid-friendly sources.

Science check: The "Travel Warnings" panel is where the standard lives — require data (temperatures, atmosphere, surface) not just "it's scary."

Grading rubric (suggested): Scientific accuracy (40%), required elements complete (25%), creativity (20%), presentation (15%).

10 Vocabulary Bingo

REVIEW GAME

OBJECTIVE
Review chapter vocabulary in a low-stress game format.

TIME
20–30 minutes

STANDARDS
S6E1 (all)

How to Play

1. Students write one word from the word bank into each square of their board, in any order. (FREE space stays free.)
2. The teacher reads a **definition** (not the word!). Students mark the square if they have the matching word.
3. First to 5 in a row calls "BINGO!" — but must read back each marked word AND its meaning to win.
4. Play again with "blackout" rules (whole board) for a longer review.

Word Bank (24 words + FREE)

universe	geocentric	planet	asteroid
galaxy	heliocentric	rocky planet	asteroid belt
Milky Way	orbit	gas giant	meteoroid
solar system	gravity	habitable zone	meteor
star	inertia	dwarf planet	meteorite
model	mass	comet	telescope

		FREE 🚀		

📘 **TEACHER NOTES**

Read definitions straight from the Vocabulary Reference sheet (Section 1) so wording matches what students studied. Keep a checklist of called words to verify BINGOs. For struggling readers, display the word bank on the board during play.